A Secure File Sharing System Based on IPFS and Blockchain

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Certificate

This is to certify that the seminar report titled "**A Secure File Sharing System Based on IPFS and Blockchain**" is a bonafide record of the work carried out by **Yadukrishnan N K (Univ.Reg.No. VAS23CS197)** of Vidya Academy of Science & Technology, Thalakkottukara, Thrissur - 680 501 in partial fulfillment of the requirements for the award of **Degree of Bachelor of Technology in Computer Science and Engineering** of **APJ Abdul Kalam Technological University**, during the academic year 2023-2027.

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Abstract

Centralized cloud storage services dominate modern file sharing, yet they concentrate data, identity and access control under a single provider, creating single points of failure, limited user ownership and weak auditability. This report presents a secure file sharing system that combines the InterPlanetary File System (IPFS) with the Ethereum blockchain to provide decentralized storage and decentralized access control. The proposed framework uses a ReactJS web client, the MetaMask wallet for authentication and transaction signing, Solidity smart contracts that record file hashes and enforce permissions on-chain, and a private IPFS swarm hosted on dedicated Digital Ocean droplets. Unlike typical Web3 applications that rely on third-party pinning services such as Pinata, the proposed system avoids public pinning entirely, thereby removing gateway dependence and reducing latency. Files are divided into 256 KB chunks, each identified by its own cryptographic hash, and distributed across five swarm nodes, while only the lightweight Content Identifier (CID) and its metadata are committed to the blockchain, keeping gas cost low. Owners can grant or revoke access for specific wallet addresses at any time through a single signed transaction, and every grant and revoke action is permanently recorded to form a tamper-proof audit trail. A working ReactJS prototype demonstrates upload, sharing, revocation and download, and the dedicated droplets achieve an average latency below 180 ms compared with 500 ms to 2 s on generic public gateways. The framework lays a practical foundation for user-controlled, resilient and auditable file sharing and can be extended toward multi-cloud backup, zero-knowledge metadata privacy and social-recovery wallets.

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Vision of the Institution

To be a premier institution in engineering education and research, nurturing innovative professionals for sustainable societal development.

Mission of the Institution

M1. Provide quality engineering education that empowers students with strong technical competence and analytical problem solving abilities.

M2. Promote research, innovation, entrepreneurship and industry collaboration to develop sustainable technological solutions.

M3. Develop ethical and socially responsible professionals with leadership qualities, lifelong learning abilities and a commitment to societal well-being.

Vision of the Department

To excel in computer science and engineering education and research, nurturing innovative and ethically responsible professionals for sustainable technological development.

Mission of the Department

M1. Impart quality education in computer science and engineering by providing strong theoretical foundations, practical skills and exposure to emerging technologies.

M2. Promote research, innovation, entrepreneurship and industry collaboration to develop sustainable and socially relevant computing solutions.

M3. Develop graduates with professional competence, ethical values, logical and analytical thinking skills, leadership qualities and a commitment to lifelong learning and societal well-being.

Program Educational Objectives (PEOs)

PEO-1: Graduates shall practice computer science and engineering in IT industry, public sector undertaking, as an entrepreneur or as a researcher pursuing a successful professional career.

PEO-2: Graduates shall demonstrate creativity, develop innovative ideas and work in teams to accomplish a common goal.

PEO-3: Graduates shall demonstrate project management skills, lifelong learning and social awareness upholding ethical values.

Program Outcomes (POs)

PO1	**Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
PO2	**Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
PO3	**Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for public health, safety, and environment. (WK5)
PO4	**Conduct Investigations of Complex Problems:** Conduct investigations using research-based knowledge including design of experiments, modelling, analysis and interpretation of data. (WK8)
PO5	**Engineering Tool Usage:** Create, select and apply appropriate techniques and modern engineering tools recognizing their limitations. (WK2 and WK6)
PO6	**The Engineer and The World:** Analyze societal and environmental impacts with reference to sustainability, economy, health, safety, and culture. (WK1, WK5, WK7)
PO7	**Ethics:** Apply ethical principles, professional ethics, and adhere to laws. (WK9)
PO8	**Individual and Collaborative Team Work:** Function effectively as an individual and in teams.

PO9	**Communication:** Communicate effectively through reports, documentation, and presentations considering diverse audiences.
PO10	**Project Management and Finance:** Apply management principles and economic decision-making in projects.
PO11	**Life-Long Learning:** Recognize the need for independent and lifelong learning. (WK8)

Knowledge and Attitude Profile (WK)

WK1	A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
WK2	Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
WK3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
WK4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
WK5	Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.
WK6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.

WK7	Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.
WK8	Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.
WK9	Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.

Program Specific Outcomes (PSOs)

PSO-1	The ability to apply the knowledge of computational structures and algorithms for solving complex engineering problems in multidisciplinary areas.
PSO-2	The ability to design, develop and evaluate computer-based systems using advanced tools and technologies to meet real-world challenges.

Sustainable Development Goals (SDGs)

SDG 1	**No Poverty:** End poverty in all its forms everywhere.
SDG 2	**Zero Hunger:** End hunger, achieve food security and improved nutrition and promote sustainable agriculture.
SDG 3	**Good Health and Well-being:** Ensure healthy lives and promote well-being for all at all ages.
SDG 4	**Quality Education:** Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.
SDG 5	**Gender Equality:** Achieve gender equality and empower all women and girls.
SDG 6	**Clean Water and Sanitation:** Ensure availability and sustainable management of water and sanitation for all.
SDG 7	**Affordable and Clean Energy:** Ensure access to affordable, reliable, sustainable and modern energy for all.
SDG 8	**Decent Work and Economic Growth:** Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all.
SDG 9	**Industry, Innovation and Infrastructure:** Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation.
SDG 10	**Reduced Inequalities:** Reduce inequality within and among countries.
SDG 11	**Sustainable Cities and Communities:** Make cities and human settlements inclusive, safe, resilient and sustainable.
SDG 12	**Responsible Consumption and Production:** Ensure sustainable consumption and production patterns.

SDG 13	**Climate Action:** Take urgent action to combat climate change and its impacts.
SDG 14	**Life Below Water:** Conserve and sustainably use the oceans, seas and marine resources for sustainable development.
SDG 15	**Life on Land:** Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss.
SDG 16	**Peace, Justice and Strong Institutions:** Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all levels.
SDG 17	**Partnerships for the Goals:** Strengthen the means of implementation and revitalize the global partnership for sustainable development.

SDGs addressed with Justification

Seminar Title	**SDG addressed**	**Justification**
Secure File Sharing using IPFS and Blockchain	SDG 9: Industry, Innovation and Infrastructure	Presents a resilient, decentralized storage infrastructure built on IPFS, Ethereum smart contracts and self-hosted swarm nodes, reducing dependence on centralized providers. This is the primary goal addressed.
	SDG 4: Quality Education	Lets students and institutions store grade cards, theses and certificates with owner-controlled access and hash-based verification of academic records.
	SDG 3: Good Health and Well-being	Allows medical documents to be shared privately with a chosen doctor's wallet and withdrawn later, protecting patient data.

POs & PSOs addressed with Justification

Seminar Title	POs & PSOs addressed	Justification
Secure File Sharing using IPFS and Blockchain	PO1 – Engineering Knowledge	Applies cryptographic hashing (SHA-256), peer-to-peer storage, distributed systems and blockchain fundamentals to secure file sharing.
	PO2 – Problem Analysis	Analyzes weaknesses of centralized storage such as single points of failure, loss of ownership and static access control, using the literature.
	PO3 – Design/Development of Solutions	Designs an architecture of ReactJS client, MetaMask, Solidity contracts and a private IPFS swarm that keeps data private and available.
	PO4 – Conduct Investigations of Complex Problems	Reviews prior work on IPFS, blockchain access control and decentralized storage and compares ownership, integrity, permissions and latency.
	PO5 – Modern Tool Usage	Uses ReactJS, MetaMask, Solidity, IPFS, Ethereum and Digital Ocean, and recognizes limits such as gas cost and private key loss.
	PO6 – The Engineer and The World	Gives users control of their data, reduces the impact of data breaches and provides an auditable trail for institutions.

Seminar Title	**POs & PSOs addressed**	**Justification**
	PO7 – Ethics	Keeps file content off-chain, records grant and revoke actions transparently and respects the owner's privacy.
	PO8 – Individual and Team Work	Works independently and incorporates feedback from the guide, coordinators and peers during the seminar.
	PO9 – Communication	Communicates the study through this report and a seminar presentation.
	PO10 – Project Management and Finance	Plans the seminar within the schedule and considers gas fees and hosting cost in the design.
	PO11 – Life-Long Learning	Studies emerging Web3 technologies independently, including IPFS, smart contracts and wallet-based identity.
	PSO1	Applies content hashing, 256 KB chunking and permission mappings to solve a storage and access problem.
	PSO2	Evaluates a working ReactJS prototype built with modern blockchain tools against real-world requirements.

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List of Abbreviations

IPFS	InterPlanetary File System
CID	Content Identifier
P2P	Peer-to-Peer
DApp	Decentralized Application
ETH	Ether (Ethereum currency)
SHA	Secure Hash Algorithm
CORS	Cross-Origin Resource Sharing
API	Application Programming Interface
ZK-SNARK	Zero-Knowledge Succinct Non-Interactive Argument of Knowledge

Chapter 1

INTRODUCTION

1.1 Background and Motivation

The volume of digital data generated by individuals, institutions and enterprises is growing explosively, and the cloud has become the default place to keep it. Student records, medical documents, theses and corporate files are routinely uploaded to centralized providers that store, index and serve them from a small number of data centers. This model is convenient, but the scale and complexity of today's digital world makes purely centralized methods increasingly inadequate and insecure. A single compromised server, a misconfigured bucket or a prolonged outage can expose or deny access to thousands of sensitive records at once.

A second concern is ownership. When a file is placed on a cloud platform, the provider effectively controls how it is stored, replicated, accessed and retained, while the user holds only an account credential. Access control is typically a static setting maintained inside the provider's own database, and the user has no independent, verifiable evidence of who accessed a file and when. Prioritizing data security therefore involves not only encryption in transit but also giving individuals and organizations real authority over how their data is stored and shared (Benet, 2014; Steichen et al., 2018).

Decentralized technologies offer an alternative. IPFS links devices through a network of nodes and locates files by their content rather than by a fixed server address, while blockchain provides an immutable, transaction-verified layer that guarantees the authenticity of interactions. Together they separate where data lives from who is allowed to see it, and this separation motivates the present work.

1.2 Technological Advancements in Decentralized Storage

Technological progress in peer-to-peer networking and distributed ledgers has changed how files can be stored and shared. In the **InterPlanetary File System (IPFS)**, every file is split into blocks and each block is addressed by a cryptographic hash, called its **Content Identifier (CID)**. A file is therefore requested by what it is rather than where it is, which improves redundancy: the content remains reachable as long as at least one node holds it, even during localized network failures or route drops (Trautwein et al., 2022).

Blockchain platforms such as Ethereum add a consensus and trust layer. Smart contracts written in Solidity execute deterministically on every node, record state permanently and can enforce rules without a central administrator. Researchers have combined the two technologies in several ways: Roychowdhury et al. used a private blockchain with IPFS for secure student file storage, Alizadeh et al. proposed efficient decentralized storage that couples a public blockchain with IPFS, and Steichen et al. explored blockchain-based decentralized access control designed specifically for IPFS content.

Wallet software such as **MetaMask** completes the stack by holding the user's private keys inside the browser and signing every authentication and transaction request. Identity thus becomes a cryptographic wallet address rather than a username stored in a provider's database, which supports the concept of self-sovereign identity.

Despite these advances, most Web3 file applications still depend on public gateways or third-party pinning services such as Pinata for availability. These introduce latency and break the data control loop, because the data owner no longer controls the nodes that actually hold the bytes. Running a self-managed private swarm closes this gap.

1.3 Role of IPFS and Blockchain in Secure File Sharing

The proposed system treats IPFS and blockchain as two coordinated layers. IPFS acts as the **storage fabric**: it holds the actual bytes of every file, chunked and distributed across swarm nodes. The blockchain acts as the **consensus and trust layer**: it records which wallet owns which CID and which other wallets are authorized to read it.

Because storing heavy files directly on-chain is impractical and prohibitively expensive, the design follows a **hash decoupling principle**. Only the lightweight CID, together with metadata such as file name, size and timestamp, is committed to the ledger, which keeps gas fees low and transactions fast. The CID still cryptographically proves file integrity whenever the file is fetched, since any change in the content changes the hash.

Access control is handled on-chain. Permissions are linked to 160-bit Ethereum wallet addresses, so the owner can allow or revoke a specific peer with a single signed transaction, and the status of every peer is visible as either *Authorized* or *Revoked*. Since every change is permanently recorded, the system also provides a complete, tamper-proof audit trail, something that conventional cloud storage and generic IPFS gateways do not offer.

1.4 Goals and Purposes

The overarching goal of this study is to build a secure, user-controlled file sharing platform that removes dependence on centralized storage vendors and third-party IPFS pinning services while still delivering competitive performance.

The first objective is to develop a standalone, browser-ready platform in which every user action is authenticated through MetaMask and Ethereum smart contracts. The second is to store files across isolated, self-hosted private IPFS swarm nodes running on dedicated Digital Ocean droplets, so that availability and latency are under the owner's control.

The third objective is dynamic sharing: users should be able to grant and revoke file access to specific peers at any time, which standard distributed storage cannot do once metadata has been posted. The fourth is file integrity assurance through content hashing, so that files cannot be tampered with undetected.

Ultimately, the purpose of this work is to offer a real and usable alternative to mainstream centralized cloud storage, demonstrated through a working prototype rather than a purely theoretical model.

Chapter 2

LITERATURE REVIEW

2.1 LITERATURE SURVEY

2.1.1 Blockchain and IPFS for Secure Student File Storage

- **Authors:** Roychowdhury et al.
- **Focus:** Secure storage of student files using IPFS and a private blockchain

Roychowdhury et al. combined blockchain and IPFS to store student files securely. Files are placed on IPFS while their hashes are recorded on a private blockchain, and access to a file is granted through hash-based access. The approach demonstrates that sensitive academic records can be kept off a central server while remaining verifiable.

The system shows the practicality of the IPFS and blockchain pairing in an institutional setting. However, a private blockchain limits openness and auditability, and the design does not address dynamic revocation of access to a file that has already been shared. Dependence on externally hosted IPFS nodes also leaves availability outside the owner's control.

2.2 LITERATURE SURVEY

2.2.1 Design and Evaluation of IPFS: A Storage Layer for the Decentralized Web

- **Authors:** Trautwein et al., 2022

- **Source:** ACM SIGCOMM

Trautwein et al. (2022) presented an in-depth evaluation of IPFS as a storage layer for the decentralized web. Their measurements span roughly 2700 autonomous systems in 152 countries and examine how content is published, discovered and retrieved through the distributed hash table and the Bitswap protocol. The study confirms that content-addressable routing works at global scale and identifies where retrieval latency arises.

The work establishes IPFS as a mature storage fabric, but it also shows that retrieval performance varies with the location and number of providers of a block. This observation supports the use of dedicated, well-connected nodes that pin content directly, which is exactly the approach adopted in the present framework. The study does not, however, address who is permitted to read the data.

2.3 LITERATURE SURVEY

2.3.1 Blockchain-based Decentralized Access Control for IPFS

- **Authors:** Steichen et al., 2018

- **Source:** IEEE International Conference on Blockchain

Steichen et al. (2018) explored a decentralized access control mechanism designed specifically for IPFS content. Because IPFS has no built-in notion of permissions, anyone who knows a CID can request the corresponding data. The authors therefore use Ethereum smart contracts to record access rights and to let an owner decide which parties may obtain a file, removing the need for a central authorization server.

The study demonstrates that on-chain access control is feasible for IPFS content. Nonetheless, challenges remain in the cost of repeated contract interaction and in how quickly a revocation takes effect in practice. These are the aspects that the present system targets through per-user permission records and one-click revoke transactions.

2.4 LITERATURE SURVEY

2.4.1 Efficient Decentralized Storage and Web 3.0 in Education

- **Authors:** Alizadeh et al.; Lal et al.
- **Focus:** Public-blockchain storage with IPFS; evolution from Web 1.0 to Web 3.0

Alizadeh et al. proposed an efficient decentralized storage scheme that combines a public blockchain with IPFS to improve reliability, keeping bulk data off-chain and only references on-chain. Lal et al. traced the evolution of the web from Web 1.0 to Web 3.0 and discussed its role in immersive, decentralized education, where learners and institutions need to keep ownership of their records.

Taken together, all of the surveyed work confirms IPFS and blockchain as a proven, tamper-resistant storage pairing. The common gap is the reliance on public gateways or pinning services, static or weak permission models, and the absence of a deployed, self-hosted swarm. The proposed system addresses these gaps in a single prototype.

Chapter 3

COMPONENTS OF THIS SYSTEM

3.1 ReactJS Web Client and MetaMask Authentication

The ReactJS front-end is the portal where users land, authenticate and manage their files. Authentication is performed through the MetaMask browser wallet: the application requests a signature, MetaMask signs it with the user's private key, and the recovered wallet address becomes the user's identity. No password or central authentication database is needed, and every upload, share or revoke request must be signed by the wallet.

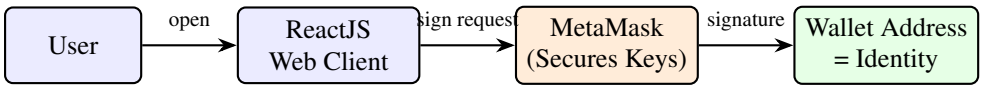


Figure 3.1: Wallet-based authentication module

3.2 Smart Contract Layer

Solidity smart contracts record the CID-to-owner mapping and enforce access permissions on-chain. The contract stores, for each upload, the IPFS hash together with the file name, size, timestamp and uploader's address. It also maintains a per-user permission table so that the owner can mark any peer wallet as authorized or revoked. Transactions execute on Ethereum and require a small ETH gas payment.

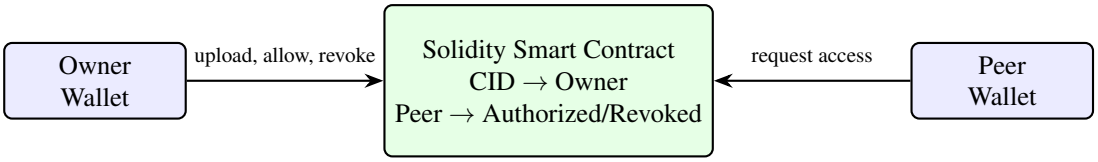


Figure 3.2: Smart contract and access control module

3.3 Private IPFS Swarm on Digital Ocean

The storage layer is a private IPFS swarm of five micro-servers hosted as Digital Ocean droplets. The droplets pin and replicate file chunks, which bypasses the speed limits of public gateways and removes dependence on third-party pinning services. Each node is configured through its IPFS `config.json`: Cross-Origin Resource Sharing (CORS) headers restrict which origins may call the API, the API listens on port 5001, the HTTP gateway on port 8080 and the swarm on TCP port 4001, which connects the nodes to one another.

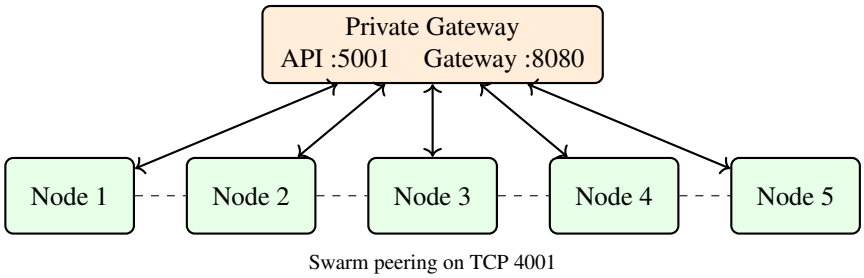


Figure 3.3: Private IPFS swarm of five droplet nodes

The relevant part of the swarm node configuration is shown below.

```
{ "API": { "HTTPHeaders": {
    "Access-Control-Allow-Origin": ["*"],
    "Access-Control-Allow-Methods": ["PUT","GET","POST"] } },
  "Addresses": {
    "API": "/ip4/143.110.246.230/tcp/5001",
    "Gateway": "/ip4/143.110.246.230/tcp/8080" } }
```

3.4 File Chunking and Hashing Module

When a file is uploaded it is divided into 256 KB binary chunks. Each chunk receives its own hash identifier so that it can be verified independently, which prevents a single

point of corruption from destroying the whole file. The chunks are spread across multiple swarm nodes rather than kept on one server. When a transaction resolves, the browser fetches the chunks, decrypts and reassembles them, and rebuilds the original file inside its own sandbox.

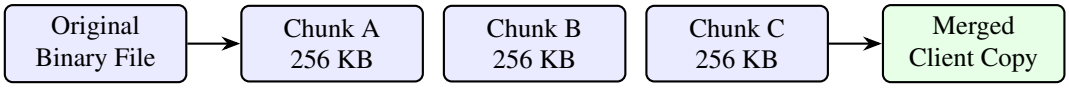


Figure 3.4: Binary partitioning and reassembly

Chapter 4

METHODOLOGY

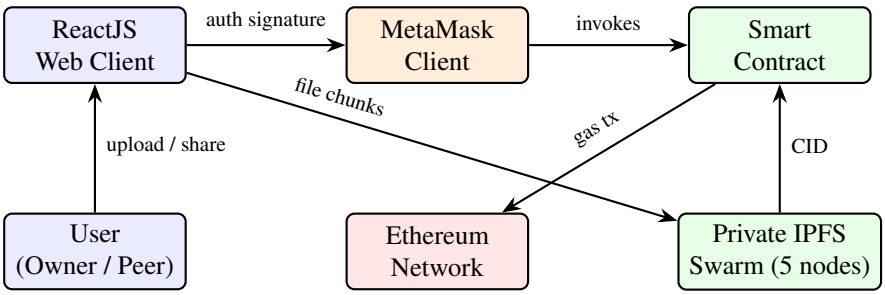


Figure 4.1: Working Flow Chart

4.1 Working Mechanism

4.1.1 Working Mechanism

The proposed system is a decentralized file sharing platform in which the browser is the central coordinating unit. The user lands on the web page and authenticates with a MetaMask signature, then chooses an action: upload, download, share or revoke. The client connects to Ethereum and pays gas to execute the transaction, while the file itself is hashed and sent to IPFS through the private gateway. Before any file is released the system verifies that only authorized wallet addresses can reach it, and finally the Solidity contract permanently records the hash-to-owner mapping on-chain.

4.1.2 User Authentication

The flow begins with MetaMask-based authentication at the start state. The wallet signs a challenge so that identity is verified before any action proceeds. Because access rights are checked client-side using wallet pairings, no centralized authentication

database is required. Once verified, the user reaches the main file management dashboard.

4.1.3 File Upload and Ledger Registration

The upload sequence consists of the following steps.

1. The user logs into the web application through MetaMask.
2. The local browser engine calculates a cryptographic hash of the file binary.
3. Data chunks are transferred securely to the self-managed private IPFS swarm.
4. IPFS issues a unique 46-character Content Identifier (CID) for the uploaded file.
5. A transaction maps the new CID to the uploader's wallet address.
6. The Solidity smart contract permanently logs this relationship on-chain.

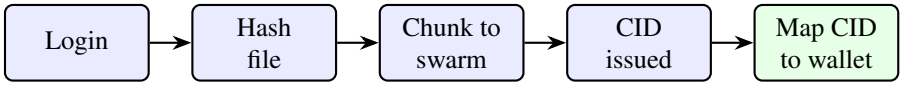


Figure 4.2: File upload and ledger registration

4.1.4 Hash Decoupling Principle

Storing heavy files directly on the blockchain is impractical, so data hosting is decoupled from ledger updates. The CID is derived from the content,

$$\mathrm{CID}_{\mathrm{file}} = \mathrm{SHA\text{-}256}(\mathrm{BinaryData})$$

and only this lightweight identifier is committed to the blockchain. Committing the CID rather than the file keeps transaction cost low and improves system speed, while the actual bytes live only on the IPFS swarm. The CID still proves file integrity whenever it is fetched.

4.1.5 Dynamic Access Control and Sharing

A file owner shares a file by entering the recipient's MetaMask wallet ID. A smart contract entry is created that grants the specified peer access to the file, and only wallets with verified blockchain credentials can decrypt shared content. To withdraw access,

the owner issues a revoke transaction on Ethereum, which immediately denies the peer's future connections. A visible peer wallet ledger tracks the state of every peer, as shown in Table 4.1.

Table 4.1: Access authorization ledger

Peer Wallet	**State**	**Command**
0x71C. . . 397a	Authorized	Revoke
0xAA9. . . b8c3	Revoked	Allow

4.1.6 File Retrieval and Reconstruction

When an authorized request arrives, the system locates the chunks across the swarm nodes, verifies each chunk by its hash, and returns them to the browser, which decrypts and reassembles the original file inside its sandbox and delivers it to the user.

4.1.7 Actors and Data Flow

The system has the following actors. The *User* signs in, stores files and manages access to personal data. The *Peer* receives shared access and retrieves files granted by other users. The *Backend or authentication layer* handles sign-in and identity verification. The *Developer* maintains the system and adds new nodes or infrastructure as required.

For an upload, the data flows as follows: the user presses the upload button, the system verifies authentication, the backend calls the decentralized storage layer, the file is hashed to produce its content identifier, the file metadata and hash are returned to refresh the file list, and the bytes finally rest across the private swarm nodes.

4.1.8 Security and Performance Comparison

The proposed swarm node solution was compared with public cloud storage and a generic IPFS gateway on four parameters, summarized in Table 4.2. The prototype reports an average latency below 180 ms through the dedicated droplets, against 500 ms to 2 s for a generic gateway.

4.1.9 Prototype Dashboard

The prototype dashboard is built on ReactJS for seamless connection management. It lists every stored file with its content hash (CIDv0) and a one-click download action, displays the active droplet IP to confirm the private gateway is online, and shows a peer

Table 4.2: Architecture evaluation matrix

Parameter	**Public Storage (AWS / GCS)**	**Generic IPFS Gateway**	**Proposed Swarm Node**
Data Ownership	Zero (provider's control)	Shared (public hosts)	Absolute (private swarm)
File Integrity	Periodic internal checks	Cryptographic hash mapping	Real-time Solidity ledger check
Permissions Access	Static account settings	No native way to revoke	Dynamic MetaMask signatures
Average Latency	$<$ 100 ms	Variable (500 ms – 2 s)	Fast ($<$ 180 ms via droplets)

access panel with Allow and Remove controls. Revoking access invalidates the peer's future connections.

Chapter 5

APPLICATION

5.1 Academic Records Management:

Institutions can store grade cards, theses and certificates on the private swarm while recording only the CID and ownership on-chain. Students decide which faculty, employers or institutions may read a document, and every access change is auditable.

5.2 Healthcare Document Sharing:

Medical reports and images can be shared with a specific doctor's wallet and withdrawn once treatment ends. The owner remains in control, and the ledger shows exactly who was authorized and when.

5.3 Enterprise and Legal Document Exchange:

Contracts and confidential corporate files require fine-grained, revocable sharing between named parties. Wallet-signed permissions and an immutable audit trail meet this need without depending on a third-party storage vendor.

5.4 Tamper-Evident Document Verification:

Because the CID is derived from the content, any alteration of a stored file changes its hash. Anyone can compare a file against the on-chain record to prove it has not been modified.

5.5 Collaborative Research Data Sharing:

Research groups can exchange large datasets among authorized collaborators. Chunked, redundant storage keeps the data available even if some nodes fail.

5.6 Self-Sovereign Identity Based Access:

Using wallets as identities eliminates centralized identity providers, so organizations can offer file services without maintaining a user database.

5.7 Replacement for Public Pinning Services:

Teams that currently depend on services such as Pinata can host their own pinned storage on dedicated droplets, gaining lower latency and full control of the data path.

Chapter 6

CHALLENGES AND LIMITATIONS

6.1 Gas Cost and Transaction Latency

Every upload, share and revoke action is an Ethereum transaction that requires a gas fee and must wait for confirmation. Frequent permission changes can therefore be costly and slower than a database update, especially when network congestion is high.

6.2 Permanence of On-Chain Data

Blockchain records are immutable, so metadata written to the ledger cannot be erased. This conflicts with the right to erasure under some privacy regulations, and file names or sizes stored on-chain may leak information.

6.3 Limits of Revocation

Revocation prevents a peer from obtaining a file in the future, but it cannot take back a copy that the peer has already downloaded. Revocation also depends on the application enforcing the contract state at retrieval time.

6.4 Key Management Risk

Authentication and ownership depend entirely on the user's private key. A lost or stolen key means loss of access or loss of control, and there is no central administrator who can reset it.

6.5 Infrastructure Dependence and Scalability

A private swarm of five droplets must be operated and maintained by the owner, and its availability is bounded by those nodes. A permissive CORS setting such as Allow-Origin set to all origins must be tightened in production. Scaling to enterprise workloads requires a larger cluster and operational effort.

6.6 Adoption and Usability

Users must install a wallet and hold ETH to pay gas, which raises the barrier compared with ordinary cloud storage. Wider adoption depends on better usability and on regulatory acceptance of blockchain-based records.

6.7 Cost and Resource Requirements

Hosting dedicated droplets, paying gas and maintaining the contracts add recurring cost. These costs must be weighed against the benefits of ownership, latency and auditability.

Chapter 7

FUTURE SCOPE AND DISCUSSION

7.1 Future Developments

7.1.1 Dynamic Multi-Cloud Storage

Storage options can be expanded across AWS and Azure to provide improved backup coverage, so that files stay available even if the private swarm becomes unreachable.

7.1.2 Custom Token Options

Custom low-overhead utility tokens could be created to make gas transactions faster and cheaper than paying in ETH on the main network.

7.1.3 Zero-Knowledge Checks

ZK-SNARK protocols can be applied to verify permissions and ownership without exposing metadata, further improving metadata privacy on the public ledger.

7.1.4 Mobile Client Support

The wallet-based dashboard can be extended to native mobile applications so that users can upload, share and revoke from their phones.

7.1.5 Larger Swarm Clusters

The swarm can be scaled beyond five micro-servers for enterprise-level workloads, with automated replication and monitoring of node health.

7.1.6 Automated Key Recovery

Social-recovery wallets can be explored to reduce the risk of lost private keys, allowing trusted guardians to restore access without a central authority.

7.1.7 Stronger Content Encryption

Client-side encryption with per-file keys shared through the smart contract would protect content even from the nodes that store it.

Chapter 8

CONCLUSION

The present study demonstrates a secure file sharing system based on IPFS and blockchain that addresses the main weaknesses of centralized cloud storage: single points of failure, loss of data ownership, dependence on third-party pinning gateways and static access control. By combining a ReactJS client, MetaMask wallet authentication, Solidity smart contracts and a private IPFS swarm on dedicated Digital Ocean droplets, the framework gives users real authority over how their data is stored and shared. Hash decoupling keeps transaction cost low, 256 KB chunking with per-chunk hashing improves resilience against corruption, and wallet-signed grant and revoke actions produce a permanent, transparent audit trail.

The working prototype shows upload, share, revoke and download end to end, with average latency below 180 ms, showing that decentralization does not have to mean slow. Future work will focus on multi-cloud backup, zero-knowledge metadata privacy, mobile clients, larger swarm clusters and social-recovery wallets, thereby extending the architecture toward more scalable and user-friendly decentralized storage.

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APPENDIX

- Reference Paper of the Seminar

  A secure file sharing system based on IPFS and blockchain

- PPT of the Seminar



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